

Relative Grading System Using UET Peshawar Formula

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1 Objective

The purpose of this CEP lab is to implement a Relative Grading System in C++ that calculates student GPA and letter grades using the official UET Peshawar grading formula. The program reads marks for multiple students, assigns initial grades, computes the class average, applies a relative grading offset to bring the class average to the UET target of 65, recalculates all grades after the shift, and identifies the class topper.

2 Theory

UET Peshawar uses an absolute grading scale combined with a relative grading adjustment. In absolute grading, each student's letter grade and GPA points are determined directly from their marks out of 100 using fixed boundaries. In relative grading, the entire class's marks are shifted uniformly so that the class average reaches a target value (typically 65 at UET Peshawar). This prevents a situation where a difficult exam causes the whole class to fail unfairly.

UET Peshawar Grading Scale

Marks Range	Letter Grade	GPA
85 and above	A	4.0
80 – 84	A-	3.7
75 – 79	B+	3.3
71 – 74	B	3.0
68 – 70	B-	2.7
64 – 67	C+	2.3
61 – 63	C	2.0
57 – 60	C-	1.7
53 – 56	D+	1.3
50 – 52	D	1.0
Below 50	F	0.0

Relative Grading Formula

Class Average = Sum of all marks ÷ Number of students

Offset = Target Average (65) – Class Average

Adjusted Marks = Original Marks + Offset (capped between 0 and 100)

The program uses a struct called **Student** to store each student's name, marks, GPA, and letter grade. An array of this struct holds all students. Separate functions handle grade assignment, relative grading, average calculation (using a pointer), and finding the class topper (using a pointer). The reference parameter (&s) in `assignGrade()` allows the function to modify the struct directly.

3 Software Used

IDE	DEV-C++ IDE
Compiler	C++ Compiler (bundled with DEV-C++)

4 Procedure

Task — Relative Grading System: UET Peshawar. Declare a structure called `Student` with members for name (char array), marks (float), GPA (float), and letter grade (char array). Write a function `assignGrade()` that uses a reference parameter to assign the correct letter grade and GPA to a student based on the UET grading scale using nested if-else. Write a function `calcAverage()` that uses a pointer parameter to return the class average. Write a function `applyRelative()` that shifts all students' marks by the difference between the target average (65) and the actual class average, then reassigns grades. Write a function `findTopper()` that uses a pointer to return the index of the student with the highest marks. In `main()`, take student count and marks as input, display grades before and after relative grading, and print the class topper.

● ● ● C++ Code — `relative_grading.cpp`

```
// =====
// CEP: Relative Grading System -- UET Peshawar Formula
// =====
#include <iostream>
#include <cstring>
using namespace std;

const int MAX = 10; // maximum students

struct Student // holds one student's data
{
    char name[50]; // student name
    float marks; // marks obtained (out of 100)
    float gpa; // calculated GPA
    char grade[3]; // letter grade e.g. A, B+
};

void assignGrade(Student &s) // assign grade by UET scale
{
    if (s.marks >= 85) { s.gpa = 4.0; strcpy(s.grade, "A" ); }
    else if (s.marks >= 80) { s.gpa = 3.7; strcpy(s.grade, "A-"); }
    else if (s.marks >= 75) { s.gpa = 3.3; strcpy(s.grade, "B+"); }
    else if (s.marks >= 71) { s.gpa = 3.0; strcpy(s.grade, "B" ); }
    else if (s.marks >= 68) { s.gpa = 2.7; strcpy(s.grade, "B-"); }
    else if (s.marks >= 64) { s.gpa = 2.3; strcpy(s.grade, "C+"); }
    else if (s.marks >= 61) { s.gpa = 2.0; strcpy(s.grade, "C" ); }
    else if (s.marks >= 57) { s.gpa = 1.7; strcpy(s.grade, "C-"); }
    else if (s.marks >= 53) { s.gpa = 1.3; strcpy(s.grade, "D+"); }
```

```
else if (s.marks >= 50) { s.gpa = 1.0; strcpy(s.grade, "D" ); }
else { s.gpa = 0.0; strcpy(s.grade, "F" ); }
}

void applyRelative(Student list[], int n, float classAvg, float targetAvg)
{
    float offset = targetAvg - classAvg; // marks shift amount
    for (int i = 0; i < n; i++)
    {
        list[i].marks += offset; // shift marks
        if (list[i].marks > 100) list[i].marks = 100; // cap at 100
        if (list[i].marks < 0 ) list[i].marks = 0; // floor at 0
        assignGrade(list[i]); // recalculate grade
    }
}

void calcAverage(Student list[], int n, float *avg) // uses pointer
{
    float sum = 0;
    for (int i = 0; i < n; i++)
        sum += list[i].marks;
    *avg = sum / n; // store result via pointer
}

void findTopper(Student list[], int n, int *index) // uses pointer
{
    *index = 0;
    for (int i = 1; i < n; i++)
        if (list[i].marks > list[*index].marks)
            *index = i; // update via pointer
}

int main()
{
    Student list[MAX];
    int n = 0;
    float targetAvg = 65.0; // UET target average

    cout << "==== UET Peshawar Relative Grading System =====> endl;
    cout << "How many students? (max " << MAX << "): ";
    cin >> n;

    if (n < 1 || n > MAX) {
        cout << "Invalid. Enter between 1 and " << MAX << endl;
        return 0;
    }

    for (int i = 0; i < n; i++) // input loop
    {
        cout << "\nStudent " << (i + 1) << ":" << endl;
        cout << " Name : ";
        cin.ignore();
        cin.getline(list[i].name, 50); // read name with spaces
        cout << " Marks: ";
        cin >> list[i].marks;
        assignGrade(list[i]); // initial grade
    }
```

```
float rawAvg = 0;
calcAverage(list, n, &rawAvg); // pointer call

cout << "\n--- Before Relative Grading ---" << endl;
cout << "Class Average: " << rawAvg << endl;
for (int i = 0; i < n; i++)
cout << list[i].name << " Marks: " << list[i].marks
<< " Grade: " << list[i].grade
<< " GPA: " << list[i].gpa << endl;

applyRelative(list, n, rawAvg, targetAvg); // apply shift

float newAvg = 0;
calcAverage(list, n, &newAvg);

cout << "\n--- After Relative Grading ---" << endl;
cout << "Adjusted Average: " << newAvg << endl;
for (int i = 0; i < n; i++)
cout << list[i].name << " Marks: " << list[i].marks
<< " Grade: " << list[i].grade
<< " GPA: " << list[i].gpa << endl;

int topIdx = 0;
findTopper(list, n, &topIdx); // pointer call
cout << "\nClass Topper: " << list[topIdx].name
<< " (Marks: " << list[topIdx].marks
<< ", GPA: " << list[topIdx].gpa << ")" << endl;

return 0;
}
```

Sample Output

● ● ● Program Output

```
===== UET Peshawar Relative Grading System =====
How many students? (max 10): 5

Student 1:
Name : Ahmad Khan
Marks: 72

Student 2:
Name : Sara Bibi
Marks: 58

Student 3:
Name : Usman Ali
Marks: 45

Student 4:
Name : Fatima Noor
Marks: 81

Student 5:
Name : Bilal Hassan
```

Marks: 63

--- Before Relative Grading ---

Class Average: 63.8

Ahmad Khan Marks: 72 Grade: B GPA: 3

Sara Bibi Marks: 58 Grade: C- GPA: 1.7

Usman Ali Marks: 45 Grade: F GPA: 0

Fatima Noor Marks: 81 Grade: A- GPA: 3.7

Bilal Hassan Marks: 63 Grade: C GPA: 2

--- After Relative Grading ---

Adjusted Average: 65

Ahmad Khan Marks: 73.2 Grade: B GPA: 3

Sara Bibi Marks: 59.2 Grade: C- GPA: 1.7

Usman Ali Marks: 46.2 Grade: F GPA: 0

Fatima Noor Marks: 82.2 Grade: A- GPA: 3.7

Bilal Hassan Marks: 64.2 Grade: C+ GPA: 2.3

Class Topper: Fatima Noor (Marks: 82.2, GPA: 3.7)

5 Conclusion

This CEP lab successfully implemented the UET Peshawar Relative Grading System in C++. A struct was used to store each student's name, marks, GPA, and letter grade together, and an array of structures held all student records. The `assignGrade()` function used a reference parameter to directly update a student's grade and GPA using the official UET grading scale implemented through nested if-else statements. The `calcAverage()` and `findTopper()` functions used pointer parameters to return their results back to `main()`. The `applyRelative()` function shifted all marks uniformly to bring the class average to the UET target of 65, then recalculated grades for all students.

The program correctly displayed results both before and after relative grading, and identified the class topper. All C++ concepts from Weeks 1 to 13 — variables, loops, if-else, functions, arrays, structs, pointers, and C-strings — were combined in this single practical program.

End of CEP Lab Report — Relative Grading System — C++ Programming — UET Peshawar Formula